

# In-Text Exercises from Ch. 5: Submanifolds, Lee

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Last Updated: September 30, 2026

**Disclaimer:** The proofs given are completed by myself. Thus, they are also reviewed by myself to ensure no logic flaws and incorrect arguments/deductions/conclusions. If you find any error in my proofs, please do not hesitate to contact me so we can discuss any potential errors and I can update the document as needed.

## Exercises

### Exercise 0.1. (Exercise 5.20)

Suppose  $M$  is a smooth manifold and  $S \subseteq M$  is an immersed submanifold. Show that every subset of  $S$  that is open in the subspace topology is also open in its given submanifold topology; and the converse is true if and only if  $S$  is embedded.

*Proof.* Let  $S \subseteq M$  be an immersed submanifold. Let  $R \subseteq S$  be open in the subspace topology, that is,  $R = U \cap S$  for some open set  $U \subseteq M$ . By assumption,

$$\iota: S \hookrightarrow M, \quad p \mapsto p \quad (\text{map between the manifold structures})$$

is a smooth immersion. Since  $\iota$  is smooth, it is continuous by Proposition 2.4. Thus  $\iota^{-1}(U) \subseteq S$  is open (in the submanifold structure). Observe that

$$\iota^{-1}(U) = \{p \in S: \iota(p) = p \in U\} = U \cap S = R.$$

Hence  $R \subseteq S$  is open in the submanifold topology of  $S$ .

For the second statement, assume that  $S \subseteq M$  is an embedded submanifold and let  $R \subseteq S$  be open in its submanifold topology. By definition of  $S$  embedded, it is a manifold in the subspace topology.

Hence,  $R \subseteq S$  open implies that there exists some open  $U \subseteq M$  such that  $R = U \cap S$ . Thus,  $R$  is open in the subspace topology of  $S$  in  $M$ .

Conversely, assume that every open subset of  $S$  in its submanifold topology is open in its subspace topology. Recall that the first statement of this exercise proved that the subspace topology is contained in the submanifold topology and the assumption of the converse is the reverse inclusion. Therefore, the submanifold topology coincides with the subspace topology (they are the same). Hence, by definition,  $S \subseteq M$  is an embedded submanifold. □

### Exercise 0.2. (Exercise 5.36)

Prove the preceding proposition.

*Proof.* We state the proposition:

**Proposition 0.1.** (Proposition 5.35) Suppose  $M$  is a smooth manifold with or without boundary,  $S \subseteq M$  is an immersed or embedded submanifold, and  $p \in S$ . A vector  $v \in T_p M$  is in  $T_p S$  if and only if there is a smooth curve  $\gamma: J \rightarrow M$  whose image is contained in  $S$ , and which is also smooth as a map into  $S$ , such that  $0 \in J$ ,  $\gamma(0) = p$ , and  $\gamma'(0) = v$ .

We now prove this. Assume that  $S \subseteq M$  is an immersed or embedded submanifold and let  $p \in S$ .

Consider the smooth immersion  $\iota: S \hookrightarrow M$ . We adopt the convention of identifying  $T_p S$  with  $d\iota_p(T_p S) \subseteq T_p M$ . Let  $v \in T_p S \subseteq T_p M$ , so by our identification, there exists  $\tilde{v} \in T_p S$  such that  $v = d\iota_p(\tilde{v})$ . Let  $\tilde{\gamma}: J \rightarrow S$  be a smooth curve for  $\tilde{v}$  in  $S$ . Define  $\gamma = \iota \circ \tilde{\gamma}: J \rightarrow M$ . Hence  $\gamma(J) = \iota(\tilde{\gamma}(J)) \subseteq S$  and  $\gamma(0) = \iota(\tilde{\gamma}(0)) = \iota(p) = p$ . Observe that  $\gamma$  is smooth since it is the composition

of smooth maps and is smooth as a map into  $S$  since its corresponding  $S$ -valued map  $\tilde{\gamma}$  is smooth. Finally, using Corollary 3.25, we compute

$$\gamma'(0) = (\iota \circ \tilde{\gamma})'(0) = d\iota_p(\tilde{v}) = v.$$

Conversely, let  $\gamma: J \rightarrow M$  be a smooth curve with  $\gamma(J) \subseteq S$ , which is also smooth as a map into  $S$  such that  $0 \in J$ ,  $\gamma(0) = p$ , and  $\gamma'(0) = v \in T_pM$ . Since  $\gamma$  is smooth as a map into  $S$ , let  $\tilde{\gamma}$  denote  $\gamma$  regarded as an  $S$ -valued map, and thus  $\tilde{\gamma}'(0) \in T_pS$ . Since  $\gamma = \iota \circ \tilde{\gamma}$ , we have

$$\gamma'(0) = d\iota_p(\tilde{\gamma}'(0)) \in d\iota_p(T_pS).$$

Therefore,  $v \in T_pS$ .

□